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Self-Healing Hydrogel Dressing with Solubilized Flavonoids for Whole Layer Regeneration of Diabetic Wound

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Hydrogels are soft, tissue-like materials that have great potential as wound dressings. However, most hydrogels are unsatisfactory in achieving whole-layer regeneration of diabetic wounds due to the complex pathological microenvironment and irregular wound shapes. Here, a glucose-responsive and self-adaptive phenylboronic acid (PBA) hydrogel solubilized strong antioxidant flavonoids (Gel-Flavonoids) is developed via dynamic borate bonds. The Gel-Flavonoids system can spontaneously confirm to the irregular wound shape and release flavonoids in a high-glucose microenvironment to effectively eliminate reactive oxygen species (ROS). The optimized Gel-Flavonoids demonstrate remarkable efficacy in inhibiting inflammation and activating fibroblasts and endothelial cells through CD36-activated lipid metabolism in macrophages and is significantly superior to commercial dressings (3M) in healing rate (> 93%, 14 days) and whole-layer regeneration effect. This study obtained a multidimensional Gel-Flavonoids system to effectively repair diabetic wounds, and reveal the underlying therapeutic mechanisms, offering a promising insight to guide the development of medical materials to treat diabetic wounds.

1. Introduction

The population of diabetes mellitus patients is projected to reach 1.31 billion (1.5% of the total population) by 2050, according to the data from World Health Organization.[1] Diabetic foot ulcers lead to over 70000 lower limb amputations each year in the United States alone, with more than half of these patients expected to die within five years. This mortality rate is significantly higher than that of many cancers, including breast and prostate cancer.[2] Achieving complete layer wound healing remains a major challenge due to the complex pathological features of diabetic wounds.[3] In these wounds, irreversible oxidative damage caused by accumulated reactive oxygen species (ROS) disrupts the metabolism and polarization patterns of macrophages.[4] This immune imbalance during the wound-healing process leads to impaired new vessel formation, inactivation of matrix remodeling, and loss of hair follicles, making diabetic wounds difficult to heal.[5] Moreover, physical movement may cause the dressing to fall off and wounds to tear. Current clinical dressings are primarily single-functional, lacking customized shape, self-healing properties, and the ability to regulate the wound microenvironment.[6] Therefore, there is an urgent

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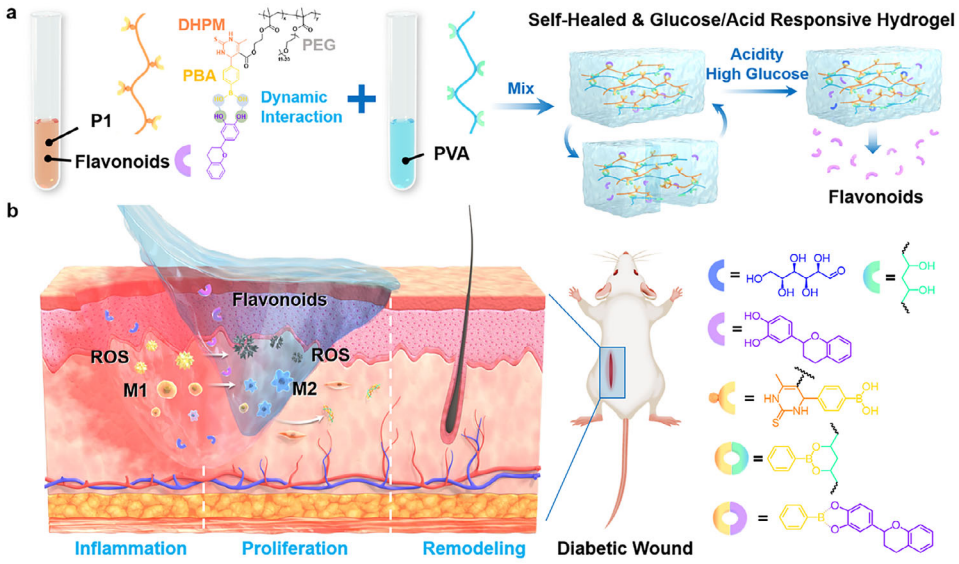
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Scheme 1. Illustration of the beneficial roles of Gel-Flavonoids for diabetic wound healing. a) Schematic illustration of the preparation procedures of Gel-Flavonoids. P1 solution with antioxidant group mixed with flavonoids to enhance the solubility, then combined with PVA solution to acquire self-healing hydrogels. Phenylboronic acid: PBA; dihydropyrimidinone: DHPM; Polyethylene Glycol: PEG. b) Mechanism of Gel-Flavonoids in promoting full-thickness chronic wound healing: In a diabetic Sprague–Dawley (SD) rat model, Gel-Flavonoids facilitate the healing of wounds from the inflammation phase to the remodeling phase. The dynamic borate bonds between flavonoids and P1 can be easily broken in high-glucose or acidic environments. Released flavonoids scavenge reactive oxygen species (ROS) to reduce oxidative stress, remodel the immune microenvironment by shifting macrophages from M1 to M2 subtype, and improve vessel and hair regeneration via immune regulation to achieve whole layer regeneration.

clinical need for an ideal dressing that can effectively modulate the complex wound microenvironment and adapt to the wound's shape.

Herb-original flavonoids are readily available and biocompatible natural products known for their potent antioxidant properties.[7] Flavonoids with o-dipheol structures such as catechin, quercetin, and myricetin have been used to treat diabetic wounds and achieved certain therapeutic effects.[8] However, the lack of efficient methods to solubilize flavonoids limits the design of biomaterials that could optimize their therapeutic potential. The direct use of flavonoids for wound treatment is often unsatisfactory due to their poor water solubility and low bioavailability.[9] Methods including usage of absorption enhancers, direct structural transformation, and glycosylation have been developed to improve the bioavailability of flavonoids. However, absorption enhancers alone can limit the dissolution rate and affect bioavailability, while directly modifying the structure may suppress the antioxidant and other biological activities of flavonoids.[10] Additionally, diabetic wounds are typically deep and irregular, with a delayed healing process.[11] This makes it challenging to maintain a long-term delivery of flavonoids on the wound surface using traditional materials that cannot spontaneously change shape to adapt to irregular wound contours.[12] Self-healing dressings can be achieved by non-covalent bonds (hydrogen bonds, van der Waals force, Guest–host interactions, and electrostatic interactions) or dynamic covalent bonds (Shiff's base, boronic ester, and disulfide). Recently, due to the stability and diversity, dynamic covalent bonds have been used to construct new self-healing hydrogels with good wound adaptability and drug-responsive release ability.[13] Therefore, by dynamic covalent bond design, the development of a multifunctional drug delivery system capable of dissolving and continuously releasing flavonoids, and spontaneously adapting to irregular shapes of wounds is urgently needed for effective diabetic wound treatment.

In this study, we report a series of adaptive phenylboronic acid (PBA) hydrogels solubilized with natural antioxidant flavonoids via a dynamic covalent bond, named Gel-Flavonoids, designed to promote diabetic wound healing with both bio-functional properties, self-adaptivity, and self-healing capacities (**Scheme 1**). Gel-Flavonoids facilitate the polarization of macrophages to M2 regenerative phenotype via lipid metabolism and the PPAR pathway. This immune microenvironment remodeling activates endothelial cells and fibroblasts, resulting in excellent, whole-layer functional regeneration of vessels, collagen, and hair follicles. In an in vivo diabetic wound model, Gel-Flavonoids effectively promoted wound healing, achieving a healing rate of over 93% within 14 days. This performance is superior to that of other groups including commercial dressings like 3M. This study demonstrates the rational design of a material system specifically for diabetic wound healing and provides new insights into the development of biomaterials for practical applications.

2. Results and Discussion

2.1. Preparation and Characterization of Gel and Gel-Flavonoids

Herb-derived flavonoids are easily accessible and biocompatible natural products with antioxidant and anti-inflammatory properties.[14] However, the lack of efficient methods to solubilize these flavonoids limits the design of biomaterials to optimize their therapeutic effects. To effectively enhance the solubility of flavonoids, the polymer (P1) was synthesized (Figure S1,

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Supporting Information). The skeleton structure of P1 includes PBA, dihydropyrimidinone (DHPM), and Polyethylene Glycol (PEG). The boric acid in P1 dynamically bonds to the o-dipheol of flavonoids and diol of poly(vinyl alcohol) (PVA) to improve the water-solubility of flavonoids and form self-adaptive hydrogel. The DHPM moiety in P1 endows the polymer and related hydrogel with antioxidant ability. The PEG segment ensures the water-solubility and biosafety of P1. Flavonoids, including catechin, quercetin, myricetin, and dihydromyricetin, were individually added to a cell culture medium (Dulbecco's Modified Eagle Medium, DMEM). Then P1 was added to evaluate the solubilization effects. Flavonoids in DMEM only served as controls (**Figure 1a**). Flavonoids in the control solution displayed minimal solubility, resulting in a turbid suspension, with the badge pattern behind the glass bottle becoming completely obscured. In contrast, the resulting P1-Flavonoid solutions were uniform and transparent, allowing the badge patterns underneath the culture dishes to be clearly visible. These results demonstrate the significant solubilization capability of P1 for flavonoids.

The antioxidant capacity of the P1-Flavonoids solution (5 mg mL^{-1} catechin, quercetin, myricetin, and dihydromyricetin in the P1 solution) was evaluated using the free radical 2,2′-azinobis(3-ethylbenzothiazoline-6-sulfonate) ($ABTS^{+\bullet}$) scavenging assay (Figure 1b). The $ABTS^{+\bullet}$ solution added with P1-flavonoids solution all faded from blue to the color of the solution itself within just 1 min compared with the PBS and P1 solution (Figure S2a,b, Supporting Information). Quantitative results further confirmed that the antioxidant performance of P1-Flavonoids was comparable to that of the standard antioxidant, water-soluble vitamin E (Trolox) (Figure S3a,b, Supporting Information). These findings indicate that the P1 solution effectively solubilized flavonoids without harming the antioxidant properties.

To accurately mimic the biophysical healing environment of tissue, hydrogels without flavonoids (referred to as Gel) were prepared by mixing P1 solution with PVA in equal volume. Similarly, by replacing the P1 solution with P1-Flavonoids solution, hydrogels with flavonoids (referred to as Gel-Flavonoids) were prepared and named Gel-Catechin (Gel-Cat), Gel-Quercetin (Gel-Que), Gel-Myrcetin (Gel-Myr) and Gel-Dihydromyricetin (Gel-2H myr), respectively. Scanning Electron Microscopy (SEM) analysis showed that the Gel possesses a porous structure, which facilitates ventilation and nutrient exchange (Figure 1c). Gel with solid content of 6, 7, and 8 wt.% exhibited a pore diameter of 10–20 µm, maintaining a structure that supports cell encapsulation. But the Gel with higher solid content (9 wt.% and 10 wt.%) showed a collapsed structure (Figure S4, Supporting Information). Small-angle X-ray scattering (SAXS) results revealed a repeating structure of 10 nm (Figure 1d). Nanoindentation curves indicated the Gel with different solid content shows the Gel possessed good flexibility, and the Gel (8 wt.%) was shown as an example (Figure 1e). And higher solid content correlating with increased hardness (Figure S5, Supporting Information). Then the strength of the Gel with different solid content (6, 7 and 8 wt.%) were evaluated. The storage modulus (G') values and loss modulus (G'') values of these hydrogels increased with increased P1 (Figure S6, Supporting Information), and Gel (8 wt.%) exhibited similar G′ and G″ of skin.[15] Therefore, a solid content of 8wt.% was used in the following experiments. Besides, Gel-Flavonoids had similar G' values (≈350 Pa) and G'' values (≈25 Pa) with those of Gel, indicating an insignificant difference to the strength by the addition of flavonoids (Figure S6b, Supporting Information). To evaluate the water absorption capacity of Gel, water absorption to the equilibrium weight of each group of hydrogels was performed according to the reported method.[16] The swelling rates of hydrogel in each group were Gel (1035%), Gel-Myr (1198%), Gel-2H myr (1199%), Gel-Que (1193%) and Gel-Cat (1258%), respectively (Figure S7a, Supporting Information). These results indicate that all the Gel and Gel-Flavonoids possess good water absorption capacity and can be used as wound dressings. The degradability has been measured by weighting remaining Gel according to reported methods of weighting remaining.[17] After 48 h, only 4.87% of the Gel remained (Figure S7b, Supporting Information). Thus, this biodegradable Gel may be used as a temporary scaffolding that supports cell growth and tissue regeneration before being replaced by natural tissue.

To meet the retention demand of wound dressings, the self-healing and self-adaptive abilities of Gel-Flavonoids were assessed. The self-healing ability was evaluated at both macroscopic and microscopic levels. For macroscopic observation, two semicircular hydrogels, one dyed blue and the other red, were separately prepared and tightly packed together (Figure 1f). After 3 h, the two pieces were fully integrated into a single disc that could not be separated by external force, demonstrating excellent self-healing ability due to the reconstruction of the network by dynamic borate ester bonds. Rheology tests quantitatively evaluated the self-healing properties of the hydrogels. The hydrogels tolerated up to 80% strain, beyond which their network structure was destroyed (Figure 1g). Further, the hydrogel network rapidly restored to its original strength after each cycle of complete destruction under 300% strain within 10 s when returned to 1% strain, confirming the outstanding self-healing ability of the hydrogels with high efficiency. (Figure 1h). Crucially, the hydrogel network was restored to its original strength after each cycle of complete destruction under 300% strain when returned to 1% strain, confirming the outstanding self-healing ability of the hydrogels. In addition, hydrogels as wound dressings also needed sufficient self-adaptability to irregular wounds. Gel-Flavonoids demonstrate good self-adaptability by confirming to different bending angles of a finger (Figure 1i). They adhered well to different substrates, showcasing their excellent retention capacity as wound dressings (Figure 1j). The hydrogel shows an adhesive strength ≈1.8 kPa, and binding flavonoids insignificantly change the adhesive strength (Figure S8c, Supporting Information). Moreover, the gel could be repeatedly reshaped into various forms, such as round, triangle, and star shapes (Figure 1k), further indicating its adaptability to irregular wounds. Together, these characteristics underscore the potential of Gel-Flavonoids for effective use in wound dressing applications.

The effectiveness of the antioxidant ability of Gel and Gel-Flavonoids was evaluated (Figure 1l). A PVA solution containing $ABTS^{+\bullet}$ (Absorbance value: ≈1.0) was mixed in equal volume with the P1-Flavonoids solution to generate the Gel-flavonoids with $ABTS^{+\bullet}$. Compared to the Gel with $ABTS^{+\bullet}$, the Gel-flavonoids with $ABTS^{+\bullet}$ rapidly faded from blue to their original color within just 1 min, indicating the complete scavenging of $ABTS^{+\bullet}$ by Gel-Flavonoids. The visual results coincided with the quantitative analysis at 734 nm (Figure S8,

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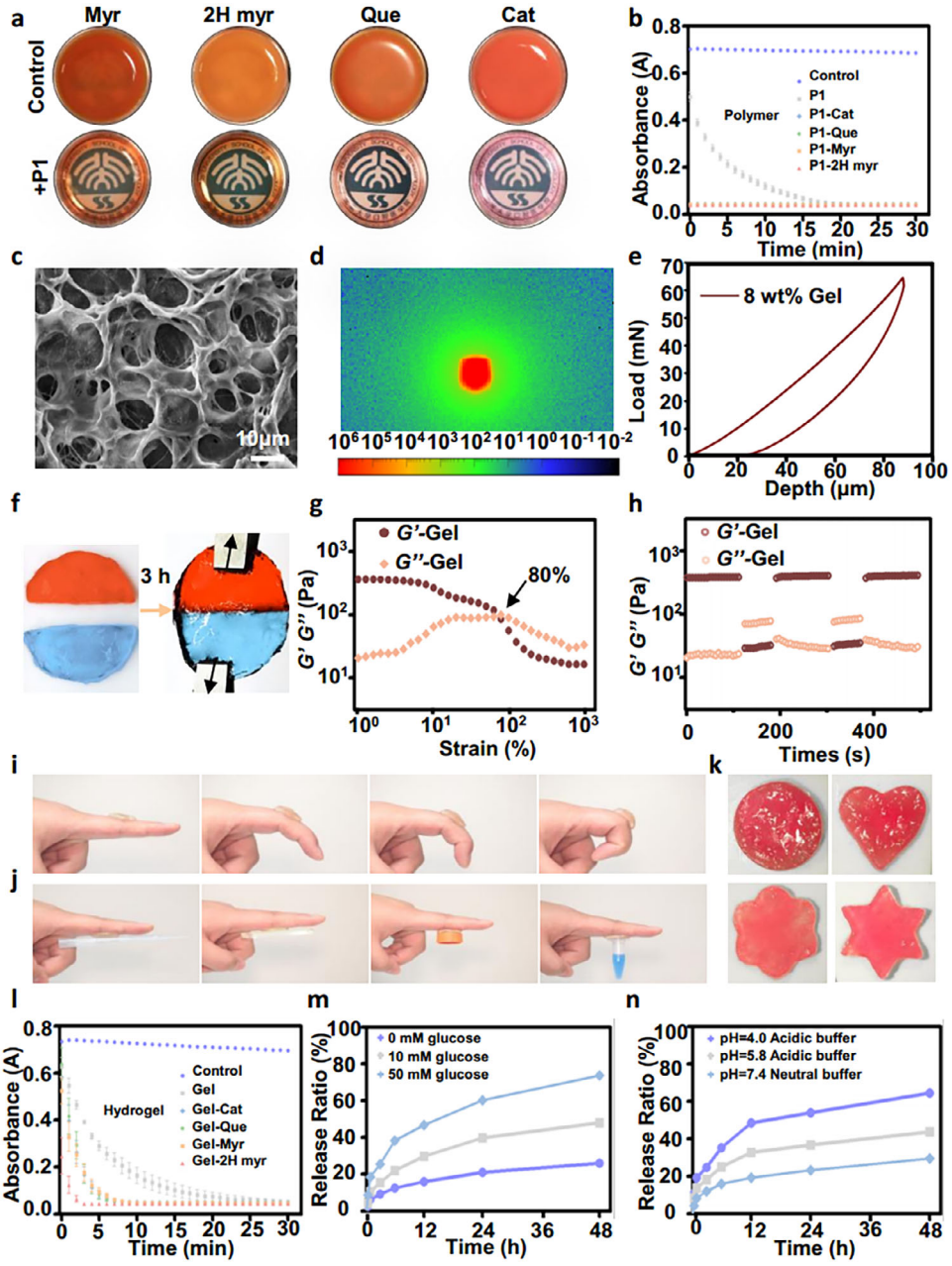


Figure 1. Characterization of polymer (P1), Gel, and Gel-Flavonoids. a) Macroscopic comparison of flavonoids (Myr, 2H myr, Que, and Cat) solubility in the control group (Dulbecco's Modified Eagle Medium, DMEM) and after adding P1. b) The absorbance of $ABTS^{+\bullet}$ in P1 and P1-Flavonoids indicating the antioxidative properties of P1 polymer and P1 with solubilized flavonoids. c) SEM images showing the porous structure of Gel (8 wt.%). d) 2D SAXS image of Gel. (e) Load and depth curves of nanoindentation for Gel (8 wt.%). f) Macroscopic self-healing property of Gel over time. g) Strain amplitude sweep test showing an intersection of the storage modulus and loss modulus of 80% in Gel. h) Rheology: alternate step strain sweep test demonstrating that the breaking and recovery behaviors of Gel are completed within 2 min at room temperature. i) Adhesion of Gel to a finger at different bending angles.(j) Adhesive capacity of Gel on different substrates. k) Shape remodeling and adaptability of Gel. l) The absorbance of $ABTS^{+\bullet}$ in Gel and Gel-Flavonoids indicating antioxidative properties of the hydrogels. m) Responsiveness of Gel-Flavonoids in different concentrations of glucose (0, 10, 50 мм): high sugar condition is conducive to the release of flavonoids. n) Responsiveness of Gel-Flavonoids in different pH conditions (pH = 4.0, 5.8, 7.4): the acidic condition is conducive to the release of flavonoids.

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Supporting Information), confirming that the Gel-Flavonoids retained excellent antioxidant performance after being prepared with P1-Flavonoids. Furthermore, the dynamic borate ester bonds impart acid and high-glucose responsive properties to Gel-Flavonoids.[18] In acidic (pH = 4.0, 5.8) and high-glucose (50 mM) environments, flavonoids were released more rapidly than under physiological conditions (pH = 7.4; glucose concentration: 10 mM) (**Figure** 2m,n). In summary, by combining P1 and PBA to solubilize flavonoids and adjusting the solid content, we successfully developed a self-healing antioxidant hydrogel with skin-like strength. This hydrogel can be easily molded into various shapes, adapting to irregular wounds, and is responsive to high-glucose environments. This makes it a promising candidate for advanced wound dressing applications, particularly for diabetic wounds.

2.2. ROS Scavenging and Immune Regulation of Gel and Gel-Flavonoids

Since the recruitment of immune cells is the initial step in wound regeneration, maintaining their viability is essential for the dynamic and complex process of diabetic wound healing.[19] Mouse-derived macrophages (Raw264.7 cells) were used to evaluate the biocompatibility of the Gel-Flavonoids system. Cell proliferation indicated that cells can proliferate well in the Gel-Flavonoids component (PVA, P1, and P1-Flavonoids) in 2D cell culture (Figure S9a, Supporting Information). Then cell proliferation and live/dead staining assays were performed on cells three-dimensionally (3D) cultured in the hydrogels. Results indicated that all the Gel-Flavonoids groups, as well as the Gel group, displayed good biocompatibility with no significant cytotoxicity, supporting their potential for further in vivo application (Figure S9b–d, Supporting Information). Hemolysis assay also showed that the hemolysis rates of red cells in Gel and Gel-Flavonoids were below 5% (Figure S9e,f, Supporting Information), further proving the biosafety of Gel-Flavonoids. In the diabetic microenvironment, the accumulation of intracellular oxidants results in excessive ROS generation, leading to immune dysregulation and chronic inflammation.[20] Given the presence of DHPM from P1 and flavonoids, Gel-Flavonoids were hypothesized to possess ROS scavenging abilities. To examine this, the Dichlorodihydro-fluorescein diacetate (DCFH-DA) probe was used to assess ROS levels in Raw264.7 cells cultured in hydrogels formed with high glucose DMEM (4.5g L^{-1}–D-glucose) and H_2O_2 (final concentration: 400 μM). The fluorescence density of DCFH-DA was significantly decreased in the Gel group, confirming the anti-oxidative property of DHPM (Figure 2a). Moreover, the cells in Gel-Flavonoids groups exhibited lower ROS fluorescence signals compared to Gel group and the positive control group, indicating the enhanced ROS scavenging ability provided by solubilized flavonoids. Therefore, Gel-Flavonoids demonstrated excellent antioxidative properties and could be used as potential applicants to improve diabetic wound healing.

Macrophages play a critical role in tissue repair and remodeling by transitioning between different phenotypes. During the early stages of wound healing, M1 polarization of macrophages induces inflammatory responses, whereas M2 polarization of macrophages facilitates tissue regeneration by regulating collagen synthesis, fibroblast activation, and vessel formation.[21] The shift from M1 to M2 dominance is crucial in the wound healing process but is significantly impaired in the diabetic microenvironment. To determine whether Gel and Gel-Flavonoids can facilitate the transition of macrophage polarization against diabetes, Raw264.7 cells were cultured in Gel and Gel-flavonoids formed with a high glucose medium. Immunofluorescence analysis revealed that the expression of CD206, an M2 marker, was significantly higher in Gel-Flavonoids compared to the Gel group (Figure 2b,c). Additionally, the fluorescence intensity of CD86, an M1 marker, was markedly reduced in Gel-Flavonoids groups relative to the Gel group. These findings suggested an enhanced M2 polarization and suppressed M1 polarization in the presence of Gel-Flavonoids. Consistent with the immunofluorescence results, RT-qPCR demonstrated that the expressions of anti-inflammatory cytokinesis (*Il10*) and M2 marker (*Arg1*) were upregulated in the Gel-Flavonoids groups (Figure 2d). Conversely, the expressions of pro-inflammatory cytokines (*Il6* and *Inos*) and M1 marker (*Ccr7*) were downregulated in the Gel-Flavonoids groups compared to the Gel group (**Figure** 3e). Among all tested groups, the Gel-Cat groups exhibited the highest expression of M2-related genes and the lowest expression level of M1-related genes, indicating the strongest ability to induce regeneration. These findings demonstrate that the Gel-Flavonoids, especially the Gel-Cat group, can enhance the M2 polarization of macrophages, thereby promoting wound healing in a high-glucose microenvironment.

To further elucidate the mechanism behind the divergent macrophage polarization effects induced by Gel and Gel-Flavonoids, RNA-sequencing analysis of Raw264.7 cells cultured in the hydrogels was conducted on day 1. Principal Component Analysis (PCA) revealed distinct heterogeneity patterns among samples from different groups (Figure S10a,b, Supporting Information). Given that the Gel-Cat group exhibited the most potent M2 induction effect, the differentially expressed genes (DEGs) between Gel-Cat and Gel groups were analyzed. Hierarchical clustering analysis demonstrated that a set of the inflammatory transcripts (*Tnfaip6, Tnfaip8, Cd80, Cox3, Cxcl9*) was downregulated in the Gel-Cat group compared to the Gel (Figure 2f). Concurrently, a set of anti-inflammatory transcripts (*Arg1, Ace, Ace2, Il10, Mrc2*) was significantly upregulated in the Gel-Cat group, reinforcing the in vitro evidence of dominant M2 polarization. Additionally, PPAR transcripts (*Ppara, Ppard*) were elevated in the Gel-Cat group (Figure 2f). KEGG pathway analysis further indicated enrichment of the PPAR signaling pathway (Figure 2g). Since PPAR activation is known to directly regulate M2 polarization of macrophages, we assumed that PPAR might play a key role in Gel-Cat-induced M2 polarization. Gene Ontology (GO) and KEGG pathway analysis also highlighted that processes related to lipid catabolism, triglyceride catabolic process, triglyceride lipase activity, and cholesterol metabolism were enriched in the Gel-Cat group (Figure 2g,h). The upregulation of these biological processes implies an increase in lipid metabolism, which is reported to be triggered by the CD36 receptor, leading to PPAR activation in macrophages.[22] Consistent with these findings, Gene Set Enrichment Analysis (GSEA) confirmed that the lipid catabolic process was enriched in the Gel-Cat group (Figure 2h). In summary, these findings indicate that Gel-Cat enhances macrophage M2 polarization, likely through the

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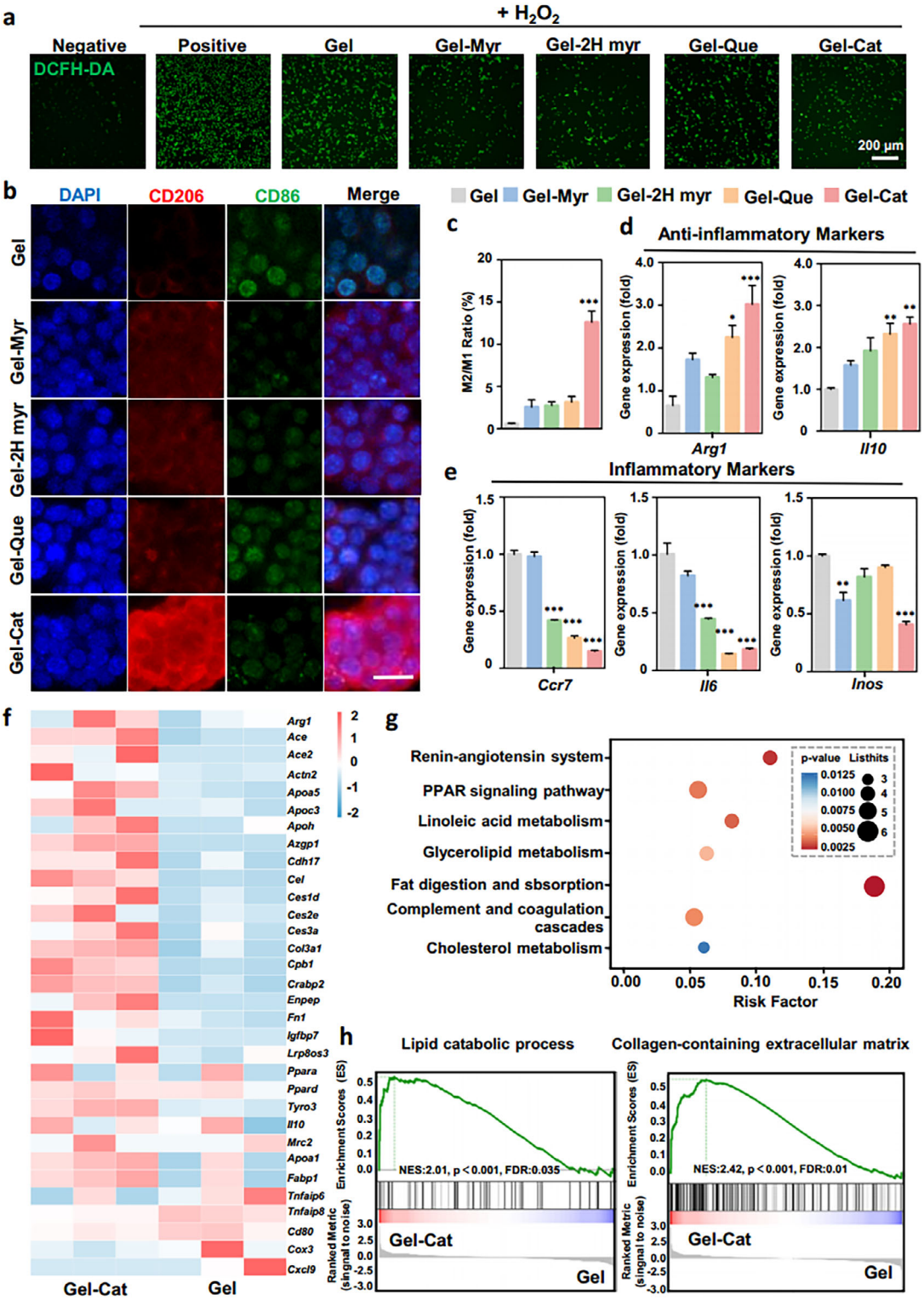


Figure 2. The immune-regulation capacity of Gel-Flavonoids in vitro. a) DCFH-DA staining showing the ROS level of macrophages in Gel and Gel-Flavonoids. b) Representative immunofluorescence images demonstrating the expression level of M2 marker CD206 and M1 marker CD86 in Gel and Gel-Flavonoids. c) Quantification of immunofluorescence images revealing that M2 (CD206+)/M1 (CD86+) cell ratios in Gel and Gel-Flavonoids. d) RT-qPCR results indicate the expression of M2 marker (*Arg1*) and anti-inflammatory factor (*Il10*) in Gel and Gel-Flavonoids. e) RT-qPCR results indicating the gene expression of M1 marker (*Ccr7*) and inflammatory factor (*Il6* and *Inos*) in Gel and Gel-Flavonoids. f) A heatmap from RNA sequencing showing the enriched M2-related and lipid metabolic transcripts in the Gel-Cat. g) KEGG analysis indicating enhanced metabolism pathways in the Gel-Cat group. h) GSEA analysis showing the enriched pathways in Gel-Cat.

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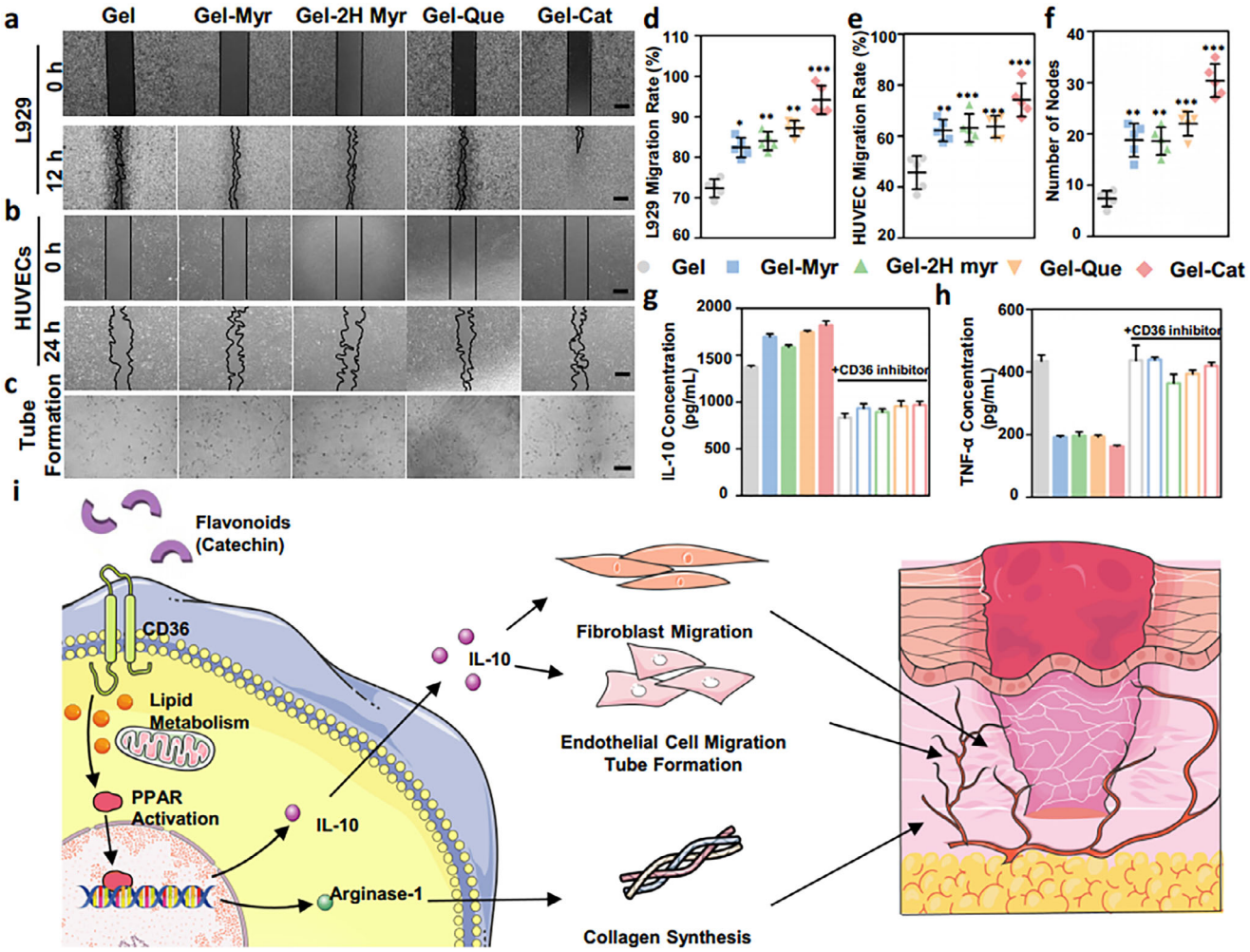


Figure 3. Gel-Flavonoids maintain the activity of HUVECs and fibroblasts. a) Scratch assay showing the migration ability of L929 cells cultured with conditional high-glucose medium from Gel-Flavonoids and Gel groups. b) Scratch assay showing the migration ability of HUVECs cultured with the conditional high-glucose medium of Gel-Flavonoids and Gel groups. c) Tube formation assay of HUVECs cultured with the conditional high-glucose medium of Gel-Flavonoids and Gel group. d) Quantification of L929 migration rate. e) Quantification of HUVECs migration rate. f) Quantification of nodes in tube formation assay. g) ELISA assay showing the secretion level of IL-10 after CD36 inhibition. h) ELISA assay showing the secretion level of TNF-α after CD36 inhibition. i) Schematic representation of the molecular signaling pathway activated by CD36, through which lipid metabolism activates PPAR and induces M2 polarization of macrophages to promote tissue regeneration.

upregulation of lipid metabolism and subsequent activation of the PPAR signaling pathway.

Importantly, genes related to extracellular matrix (ECM) remodeling were also enriched in the Gel-Cat group (Figure 2g). GO and KEGG analyses indicated an enrichment of terms and pathways associated with the extracellular space, extracellular region, and collagen-containing ECM in the Gel-Cat group (Figure 2h and Figure S10c, Supporting Information). Additionally, GSEA analysis confirmed the upregulation of collagen-containing ECM components in the Gel-Cat group (Figure 3h). These findings suggest that Gel-Cat can assist in diabetic wound healing by promoting macrophage-mediated extracellular matrix remodeling. Moreover, the enriched renin-angiotensin system pathways in Gel-Flavonoids imply an enhancement of the angiogenesis process (Figure 2h and Figure S10c, Supporting Information). In conclusion, the RNA-sequencing results demonstrate that Gel-Cat can promote wound regeneration through immune regulation, new vessel formation, and extracellular matrix remodeling.

2.3. Fibroblasts and Endothelial Cells Regulation of Gel and Gel-Flavonoids

The RNA sequencing results suggested that Gel-Flavonoids might influence wound healing through revascularization and matrix remodeling processes regulated by the immune response. To validate this, Raw264.7 macrophages were cultured in Gel and Gel-Flavonoid hydrogels with a high-glucose medium to generate a conditioned medium (CM), which was then used to assess the behavior of fibroblasts and endothelial cells. Fibroblasts are essential for tissue repair as they produce and organize the

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extracellular matrix and promote wound contraction.[23] Impaired plasticity of fibroblast leads to deficiencies of ECM production and delays wound closure. To evaluate the migration ability of fibroblasts in a diabetic microenvironment, a scratch assay was performed using the CM. The results showed that CM from Gel-Flavonoids significantly simulated fibroblast migration, with the Gel-Cat group displaying the best wound healing ability, achieving a healing rate of 95.81±3.12% (**Figure** 3a,d). Maintaining endothelial activity through immune regulation is also crucial for improving blood supply and facilitating wound closure.[24] The scratch assay indicated that complete scratch closure was achieved under CM from the Gel-Cat after 24 h, whereas scratches remained partially open in the Gel-Myr, Gel-2H myr, Gel-Que, and Gel groups (Figure 3b,e). Consequently, the chemotactic response of endothelial cells was significantly enhanced in the Gel-Flavonoids groups, particularly in the Gel-Cat group. The tube formation assay further demonstrated that more nodes, tubes, and junctions formed in Gel-Cat group, indicating superior angiogenic capabilities (Figure 3c,f and S11a,b, Supporting Information). Collectively, these observations confirmed that Gel-Cat possessed the best angiogenic capacity of HUVECs among all the Gel-Flavonoids tested. CD36 is reported to facilitate lipid metabolism in macrophages.[25] Inhibiting CD36 resulted in a decrease in IL-10 secretion by macrophage and an increase in TNF-α levels (Figure 3g,h). Additionally, the migration and activity of L929 fibroblasts and HUVECs were significantly reduced when using CM obtained with a CD36 inhibitor (Sulfo-N-succinimidyl oleate, SSO) (Figure S12a,b, Supporting Information). Thus, by activating the CD36 receptor, Gel-Flavonoids upregulated lipid metabolism and PPAR signaling, induced M2 polarization of macrophage, and increased IL-10 secretion. This, in turn, activated fibroblasts and endothelial cells, thereby enhancing the microenvironment of the diabetic wound area and promoting wound healing (Figure 3i).

2.4. In Vivo Diabetic Wound Healing Evaluation of Gel and Gel-Flavonoids

After determining that Gel-Flavonoids can affect macrophages, fibroblasts, and endothelial cells in vitro, we next investigate the regeneration efficiency of Gel-Flavonoids in vivo. A rat diabetic wound model was established by injecting Sprague–Dawley (SD) rats with streptozotocin (STZ) to induce diabetes mellitus (**Figure** 4d).[26] Blood glucose was measured one week post-injection, and rats with blood glucose levels over 16.7 mmol L^{-1} were selected for the in vivo experiments (Table S2, Supporting Information). A full-thickness wound (8 mm) was created on the dorsal skin of these rats. Comparing designed dressing with commercial dressing is the standard for demonstrating the advantages of healing effects.[27] Thus, the non-treatment control (NC), commercial dressing 3M, and non-self-healing gel GelMA were used as controls. Systematic monitoring and photographing of the wounds were performed on days 0, 3, 7, and 14. Macroscopically, all the Gel-Flavonoids treated groups exhibited faster wound closure rates compared with the control treatments (Figure 4b). Among these, the Gel-Cat group was chosen for detailed analysis due to its exemplary performance. After 14 days, significant wound areas remained in the NC (19.18±7.07%), 3M (16.46±10.43%), and GelMA (20.65±7.76%) groups, while the Gel-Cat group's wounds were almost entirely healed (wound closure rate: 95.00±3.65%) (Figure 4g). These results demonstrated that all Gel-Flavonoids significantly improved wound healing rates compared to the control groups, with Gel- Cat showing the most superior wound healing efficacy among all treatments tested. Thus, Gel-Flavonoids promoted diabetic wound healing, and Gel-Cat stands out as the most effective formulation.

Wound repair proceeds through three overlapping stages: inflammation, proliferation, and remodeling.[28] To better understand the positive effects of Gel-Flavonoids, particularly Gel-Cat, on wound healing, we conducted immunohistochemical (IHC) staining, hematoxylin and eosin (H&E) staining, and Masson's staining at different stages. During the inflammation phase, IL-10, an anti-inflammatory cytokine produced by M2 plays a crucial role in enhancing the proliferation phase.[29] The IHC staining results showed a significant increase in IL-10 expression on day 3 in Gel-Flavonoids groups, especially Gel-Cat group. This suggests that the Gel-Flavonoids groups accelerated the transition from the inflammatory phase to the proliferation phase (Figure 4c,h and S13a,b, Supporting Information). Vascularization, a critical step in the proliferation phase, supplies essential nutrients and oxygen to the wound area.[30] CD31 represents the new blood vessel formation in the wound areas. IHC staining revealed a marked increase in CD31 expression in the Gel-Flavonoids groups, with the strongest signals observed in the Gel-Cat group (Figure 4d,i and S14a, S14b, Supporting Information). H&E staining on day 14 revealed non-healing scar tissue in the NC group (Figure 4f and S15, Supporting Information). While the 3M group showed thicker tissue, it remained unhealed. In contrast, the Gel-Flavonoids groups exhibited abundant and thicker granulation tissue with shorter unhealed wound bed lengths. The Gel-Cat group had the highest number of newly-formed vessels and hair follicles (Figure 4j,k). Masson's staining demonstrated functional wound healing through collagen deposition, new vessel formation, and hair follicle regeneration. On day 14, Masson staining of the NC, 3M, and Gel groups showed dense, linearly aligned ECM fibers without secondary elements, indicating fibrotic, non-functional scarring (Figure 4e). Conversely, the Gel-Flavonoids groups showed regenerative wound healing with regrown hair follicles and glands, and less dense, more randomly oriented ECM fibers.[31] In conclusion, Gel-Flavonoids significantly enhance diabetic wound healing, with Gel-Cat showing the most substantial ability to induce regeneration.

3. Conclusion

In summary, to address the imbalanced immune response in diabetes mellitus by regulating energy metabolism, we developed a new self-healing hydrogel system named Gel-Flavonoids with a series of solubilized flavonoids. This system effectively scavenges excessive ROS, shifts macrophage polarization toward the M2 phenotype, and enhances endothelial activity and fibroblast mobility in vitro. Notably, we discovered for the first time that Gel-Flavonoids maintain immune homeostasis by regulating the lipid metabolism via CD36 receptor and activating downstream PPAR pathways. In vivo, the Gel-Flavonoids system enhances anti-inflammatory responses, promotes blood vessel formation, facilitates collagen deposition, and simulates hair regeneration,

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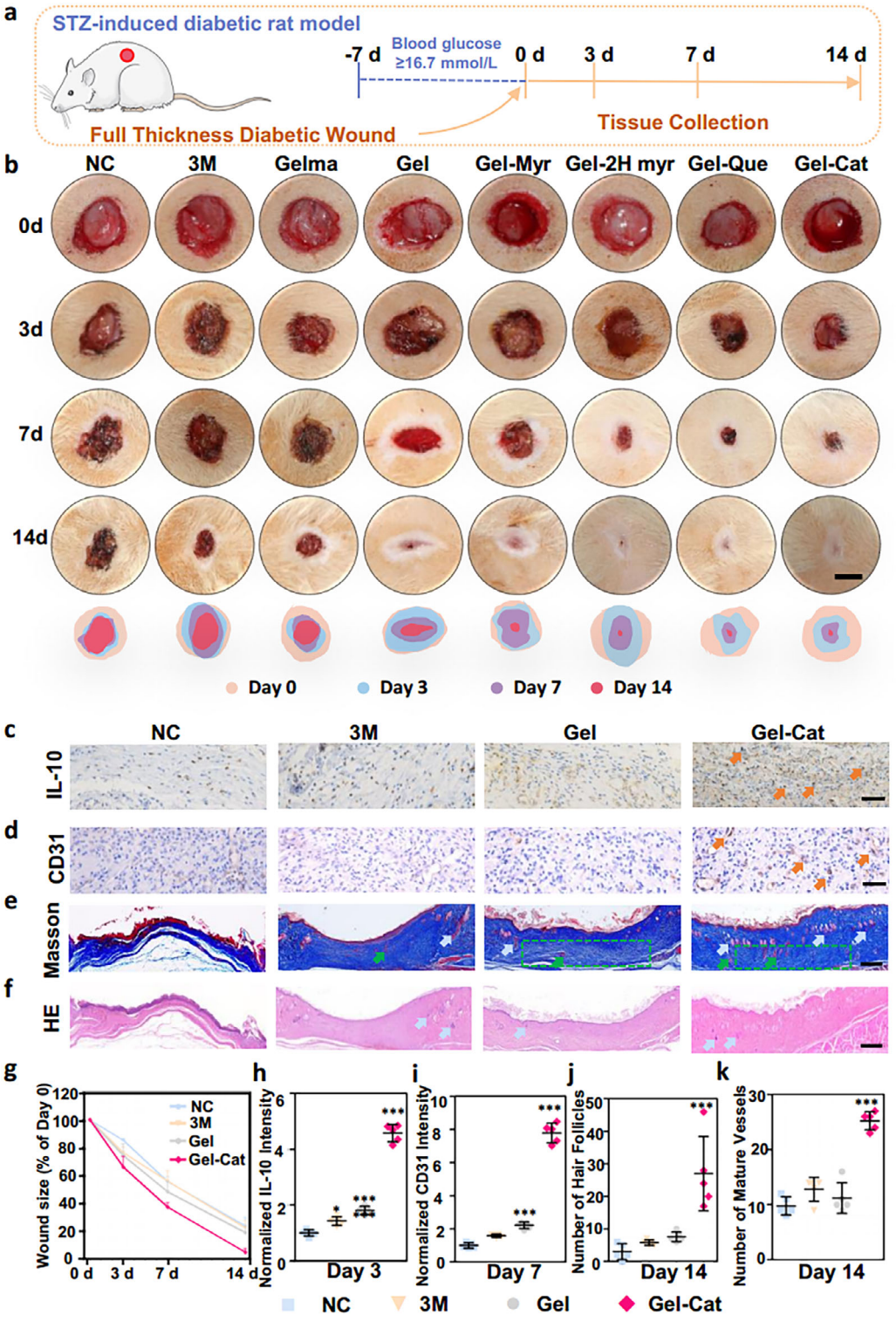


Figure 4. Gel-Flavonoids promoting the healing of full-thickness chronic wounds on the back of streptozotocin (STZ)-induced diabetic Sprague-Dawley (SD) rats. a) Schematic illustration of establishment and treatment timeline for the diabetic wound. b) Representative photographs of the wounds with different treatments. Scale bar: 5mm. c) Immunohistochemical staining of IL-10 at day 3. Scale bar: 200μm. d) Immunohistochemical staining of CD31 at day 7. Scale bar: 200μm. e) H&E staining at day 14. Scale bar: 500μm. f) Masson staining on day 14. Scale bar: 500μm. g) Quantification of the wound closure rate. h) normalized IL-10 density from immunohistochemistry staining. i) Normalized CD31 density from immunohistochemistry staining. j) Quantification of hair follicles at day 14. k) Quantification of mature vessels at day 14. Orange arrows show the immunohistochemistry stain. Light blue arrows indicate areas of nascent hair follicles, and green arrows indicate newly formed blood vessels. Green dotted rectangles indicate regenerating collagen fibers.

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resulting in promoted whole-layer functional regeneration of diabetic wounds, which has been barely reported by previous literatures. The combined in vitro and in vivo assays demonstrate that Gel-Flavonoids significantly improve wound healing. Overall, the Gel-Flavonoids system, particularly Gel-Cat, displays as a promising dressing strategy for enhancing diabetic wound healing.

4. Experimental Section

Preparation of P1 Polymer: The polymer P1 was synthesized by mixing (4-(5-((2-(Methacryloyloxy) ethoxy) carbonyl)-6-methyl-2-thioxo-1,2,3,4-tetrahydropyrimidin-4-yl)phenyl)boronic acid (M1, 96%, J&K, 20.2 g, 50 mmol), PEGMA (47.5 g, 50 mmol) and ABVN (0.25 g, 1mmol) in DMF (50 mL) in a 250 mL round-bottom flask. The flask was sealed with a rubber septum and purged by nitrogen flow for 30 min. The mixture was kept in a 70 °C oil bath for 24 h. The mixture was precipitated in diethyl ether for 3 times, and dried under vacuum to obtain the pure polymer (65 g, yield ≈96%).

Preparation of Gel-Flavonoids: To prepare the Gel- Flavonoids, a series of P1 solutions (6, 7, 8, 9, and 10 wt.% in PBS solution) were prepared by dissolving P1 (0.6, 0.7, 0.8, 0.9, 1.0 g) in PBS solution (9.4, 9.3, 9.2, 9.1, and 9.0 g, pH≈7.4) respectively. As a representative example, Gel-8 was formulated by mixing the P1 solution (8 wt.%) with a PVA solution (7 wt.% PVA in PBS solution) at a volume ratio of 1:1, and a hydrogel was quickly formed in ≈30 s after vortex (25 °C). Gel-6, Gel-7, Gel-9, and Gel-10 were prepared similarly. P1-Flavonoid solutions were prepared by dissolving 5 mg of each flavonoid (catechin, quercetin, myricetin, and dihydromyricetin) in the P1 solution (8 wt.%, 10 mL), respectively. The hydrogels with flavonoids (Gel-Flavonoids) were prepared by mixing the P1-Flavonoid solutions with an equal volume of the PVA solution. A control hydrogel without flavonoids (Gel) was prepared in the same manner, substituting the P1-Flavonoids solution with the P1 solution (8 wt.%).

Solubility: P1 solution (8 wt.%) was used to confirm the solubility of flavonoids. PBS solution served as control. After adding 1 mg flavonoids (catechin, quercetin, myricetin, and dihydromyricetin) into 2 mL P1 or PBS solution in 35mm dish, and macroscopic images were documented through the dish. Flavonoids including Quercetin, myricetin, dihydromyricetin, and catechins were added to PBS and P1 solution (8 wt.% in PBS) in centrifuge tubes with an increasing concentration from 0.1 to 10 mg mL^{-1}, respectively. The mixtures were stirred at 40 °C for 24 h, then centrifuged at 5000 rpm for 10 min. The maximum concentrations at which no precipitations occurred in the centrifuge tubes were recorded as the maximum solubility of flavonoids in PBS or P1 solution.

Scanning Electron Microscopy (SEM): The microstructures of Gel and Gel- Flavonoids were visualized by SEM (QUANTA FEG250, FEI, USA). The hydrogels were dehydrated in ethanol (30, 50, 70, 90, 100 v/v%). Dehydrated samples were put onto double-sided carbon tape, lined with silver glue, and coated with gold. Images were captured under high vacuum using at a beam power of 20 kV and a probe current of 30 µA.

Nanoindentation: The mechanical strength of all samples was determined by dividing the load by the area of the fractural cross-section. The average and maximum Young's modulus were calculated from the slope of the stress–strain curve's linear region. The toughness was defined as the amount of energy the layered film absorbed before it failed under tensile testing. Micromechanical properties of all samples were measured by a commercial TI 950 triboindenter (Hysitron) equipped with a Berkovich diamond tip (R ≈ 100nm) using a continuous depth-sensing indentation technique under ambient condition (25 °C, 30% RH).

Small-angle X-ray Scattering (SAXS) Measurement of Gel: SAXS measurements were conducted on a Xenocs Xeuss 3.0 SAXS/WAXS system using an incident Cu-Kα X-ray beam aligned parallel to the film plane. The sample-to-detector distance was maintained at 250 cm, and scattering patterns were collected with a Pilatus 300k detector. The degree of alignment of the Gel was quantified using Herman's orientation factor (f), calculated from azimuthal angle plots obtained from the 2D SAXS images. The f value, which ranges from 0 (isotropic structure) to 1 (perfect orientation), was quantified using the following equation:

$$f = \frac{1}{2}\left(3\left\langle \cos^2\varphi \right\rangle - 1\right) \tag{1}$$

where $<\cos^2\varphi>$ is the average value of the square of the cosine of the azimuthal angle for the 002 peak of the Gel. The average value was calculated as:

$$\cos^2\varphi \geq \frac{\int_0^{2/\pi} I(\varphi)\cos^2\varphi \sin\varphi d\varphi}{\int_0^{2/\pi} I(\varphi)\sin\varphi d\varphi} \tag{2}$$

A Maier–Saupe distribution function was employed to fit the φ plots as follows:

$$(\varphi) = I_0 + A\exp\left[w\cos^2(\varphi - \varphi_0)\right] \tag{3}$$

where I_0 represents the baseline intensity, φ_0 is the azimuth angle, and w is a parameter defining the width of the distribution.

Macroscopic Self-healing Ability: The self-healing properties were evaluated macroscopically. Gel-8 was selected as a representative example to demonstrate the self-healing ability and self-adaptability of these hydrogels. Two pieces of semi-circular hydrogels dyed blue and red were separately prepared and tightly packed together. Photos were taken to monitor the healing process of the two parts. Second, the recovery ability of the Gel was estimated by adhesion to the finger with different bending angles of the Gel.

Rheology: For rheological analysis, ≈500 µL of hydrogel (≈20 mm in diameter) was placed on the measuring plate of a rheometer. The storage modulus (G') and loss modulus (G'') of the hydrogel were recorded under frequency sweep step (strain = 1%, 25 °C). All hydrogels were tested through the same process.

Rheology analyses were further carried out to test the shear-thinning properties of the hydrogels. The G' and G'' values of Gel under different strains were tested (frequency = 1 Hz). Then, alternative changed strains (1%, 300%) under the same frequency (1.0 Hz) were applied to the Gel. G' and G'' values were recorded.

Adaptability: To examine the self-adaptability of the hydrogels, approximately 3 mL of Gel-8 (dyed pink) was injected through a 22-gauge needle into a circular mold. After 6 h, the hydrogel completely self-healed, conforming to the circular shape of the mold. The hydrogel was subsequently re-injected into molds of different shapes, including heart, wreath, and hexagram shapes, and images were taken to document the adaptability process.

Antioxidative Ability of P1 and Gel: The $ABTS^{+\bullet}$ solution was prepared by mixing aqueous solutions of 2,2′-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid) diammonium salt (ABTS, 3.84 mg mL^{-1}, 10 mL) and potassium persulfate (0.66 mg mL^{-1}, 10 mL), allowing it to stand overnight. A P1 solution (8% w/v in PBS) and a P1-flavonoids solution (0.5 mg mL^{-1} flavonoids in P1 solution) were prepared for the assays. The $ABTS^{+\bullet}$ solution (190 µL) was added to a 96-well plate, followed by 10 µL of the P1-flavonoids solution. The color change of the $ABTS^{+\bullet}$ solution was recorded at different time intervals. Absorbance was measured at 734 nm using a microplate reader. The P1 solution served as the control, and PBS was used as a blank. Results were expressed as mean ± standard deviation (SD), n = 3.

For further testing, a PVA solution (10% w/v in PBS, 400 µL) was combined with the $ABTS^{+\bullet}$ solution (172 µL) to create a PVA-$ABTS^{+\bullet}$ solution (7% w/v, absorbance ≈3, pH ≈7.4). The P1 and P1-flavonoids solutions were prepared as described previously. The P1-flavonoids solution (400 µL) was mixed with the PVA-$ABTS^{+\bullet}$ solution (400 µL) to form

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Gel-flavonoids with ABTS+●. A control Gel with $ABTS^{+\bullet}$ was prepared similarly, and PBS (400 µL) was combined with the PVA-$ABTS^{+\bullet}$ solution (400 µL) as a blank. Color changes were monitored, and absorbance measurements were recorded at 734 nm using a microplate reader, expressed as mean ± SD, n = 3.

In vitro pH and Glucose Response Drug Release Test: Catechin served as the model compound for evaluating the sustained-release properties of the phenylboronic acid hydrogel. A 3 mL hydrogel was prepared within a dialysis bag, and the drug was encapsulated in situ once the gel was fully formed. The bag was placed in a beaker containing 30 mL of PBS at pH values of 4.0, 5.8, and 7.4, and was maintained at room temperature. The solution was periodically collected and replenished with fresh PBS. Catechin release from the hydrogel was quantified with a UV–vis spectrophotometer, using the principal absorption peak at 280 nm.

Cell Culture: RAW264.7 macrophages were cultured at 37 °C in Dulbecco's Modified Eagle's Medium (DMEM, Gibco) supplemented with 10% fetal bovine serum (FBS, Gibco), 100 mg mL^{-1} penicillin, and 100 mg mL^{-1} streptomycin in a 5% CO_2 atmosphere. Cells were resuspended in a PVA solution (7% w/v PVA in DMEM) at a density of 2 million cells/mL. An 8% P1 solution in DMEM was prepared for solubilizing flavonoids. Equal volumes of both solutions were mixed at room temperature to achieve Gel-Flavonoids, which were then incubated under standard conditions (37 °C, 5% CO_2).

Cell Proliferation Assay: RAW264.7 macrophages were cultured in hydrogel for 6, 24, 48, and 72h, respectively. Cell Counting Kit-8 (CCK-8, Beyotime) was used. After adding CCK-8 solution into the wells, cells were incubated at 37 °C with 5% CO_2 for 2 h. Then, the supernantant was collected and placed in a 96-well plate, and absorbance at 450 nm was recorded for each sample using a microplate reader. Relative cell viability of different time points was calculated.

Live/Dead Assay: The cell viability post-24 h was assessed using a fluorescence live/dead viability assay kit (KeyGEN Bio TECH). A staining solution was prepared by dissolving 2 µ;M calcein (AM) and 8 µ;M propidium iodide (PI) in 10 mL of PBS. Cells were incubated with the staining solution for 30 min, then washed three times with PBS, after which they were observed under a fluorescence microscope (Leica). Live cells appeared green, while dead cells were stained red.

Hemolysis Assay: Fresh rat blood was collected from the abdominal aorta and treated with Heparin to prevent coagulation. The erythrocytes were subsequently separated by centrifugation at 10000 rpm for 5 min and washed with PBS until the supernatant was clear and colorless. Following this, the experiment was divided into six groups: PBS (negative control), sterile deionized water (positive control), Gel, Gel-Myr, Gel-2H myr, Gel-Que, Gel-Cat, which were mixed with red blood cell solutions at the same concentration. The hydrogel extracts were prepared in advance for testing. After incubating the mixture at room temperature for 4 h, it was centrifuged at 10000 rpm for 5 min. The absorbance of the supernatant was measured at 540 nm using a microplate reader. The hemolysis ratio was calculated with the following formula:

$$\text{Hemolysis ratio} = \frac{(\text{ODt} - \text{ODn})}{(\text{ODp} - \text{ODn})} \times 100\% \quad (4)$$

ODt: OD value of tested samples,

ODn: OD value of negative control, ODp: OD value of positive control.

Macrophage Phenotype Regulation of the Hydrogels In Vitro: RT-qPCR analysis: Total RNA was extracted from RAW264.7 encapsulated in hydrogels using TRIzol reagent (Invitrogen). Subsequently, cDNA was synthesized through reverse transcription using a PCR thermal cycler (Takara). The resulting cDNA was then amplified by RT-qPCR using TB Green Premix Ex Taq (Takara). PCR amplification was carried out with the following cycling parameters: 30 s at 95 °C, followed by 40 cycles of 5 s at 95 °C and 34 s at 60 °C. The primer sequences used to amplify *Ccr7*, *Il6*, *Inos*, *Il10*, *Arg1*, and *Gapdh* are listed in Table S1 (Supporting Information).

Immunofluorescence Analysis: Samples were rinsed with phosphate-buffered saline (PBS) and fixed with 4% paraformaldehyde for 30 min. Following fixation, samples were permeabilized with 0.1% Triton X-100 for 10 min and subsequently blocked with 3% bovine serum albumin for 1 h at room temperature. After blocking, samples were rinsed with PBS for 5 min. The following primary antibodies were applied: rabbit polyclonal antibody against CD206 (1:200, Abcam) and rat monoclonal antibody against CD86 (1:200, Abcam). After overnight incubation at 4 °C, an Alexa Fluor 647-labeled goat anti-rabbit IgG secondary antibody (1:200, Abcam) was applied for CD206, and an Alexa Fluor 488-labeled goat anti-rat IgG secondary antibody (1:200, Abcam) was used for CD86, both incubated in the dark. To visualize cell nuclei, samples were counterstained with DAPI (Solarbio). Finally, samples were observed using confocal laser scanning microscopy (Leica).

Whole Transcriptome Sequencing (RNA-Seq) and Bioinformatic Analysis: The mRNA sequencing: Whole transcriptome sequencing (RNA-Seq) analysis was performed to analyze the RNA of RAW264.7 in hydrogels. The hydrogel containing macrophages was frozen in liquid nitrogen. After adding the TRIzol solution (Invitrogen), the samples were thoroughly ground with a glass grinder. The RNA purity and quantification were assessed using a NanoDrop 2000 spectrophotometer (Thermo Scientific). RNA integrity was assessed using an Agilent 2100 Bioanalyzer (Agilent Technologies). The libraries were then built using the VAHTS Universal V6 RNA-seq Library Preparation Kit according to the manufacturer's instructions. Transcriptome sequencing was performed by OE Biotech (Shanghai, China). The Illumina Novaseq 6000 platform was used to sequence the libraries. And 150bp of paired-end reads were generated. Fastp[1] was used to manipulate the raw reads in fastq format, removing low-quality reads to obtain clean reads. HISAT22 was applied to map the clean reads to the reference genome. Calculate FPKM3 for each gene and obtain read counts for each gene using HTSeq-count4. PCA analysis was subjected to R (version 3.2.0) in order to assess the biological reproducibility of the samples. Differential expression analysis was performed using the DESeq2. Fold change > 2 or < 0.5 and p value < 0.05 was set as the threshold for significantly differential expression genes (DEGs). Hierarchical clustering analysis of DEGs was carried out using R (v 3.2.0) to show the expression pattern of genes in different groups. DEGs were enriched using R (v 3.2.0) for GO and KEGG pathways based on the hypergeometric distribution. Histograms and bubble plots of significantly enriched terms were plotted using R (v 3.2.0). Gene Set Enrichment Analysis (GSEA) was performed using GSEA software. Gene Set Enrichment Analysis (GSEA) was performed using GSEA software. The analysis used a predefined set of genes. And the genes were sorted according to their degree of differential expression in different groups.

Bioinformatics Analysis: GO and KEGG Pathway enrichment analysis of differentially expressed genes based on the hypergeometric distribution algorithm was used to screen significantly enriched functional entries. R (v 3.2.0) was used to create the column diagram and bubble diagram of significantly enriched functional items. Gene set enrichment analysis was performed using GSEA software. Predefined gene sets ranked genes according to the degree of different expression in the two types of samples. Subsequently, it was verified whether the predefined gene sets were enriched at the top or bottom of this sorted list. Data generated during this study have been deposited in Gene Expression Omnibus (GEO) with the accession code GSE260761.

Cell Migration: Following 24 h of culture in hydrogels, the medium above the hydrogels was collected. Cell debris was removed by centrifugation and filtered through a 0.22 µm filter, and the filtered conditioned medium (CM) was stored at -80 °C until use. L929 and human umbilical vein endothelial cells (HUVECs) were seeded into 6-well plates at a density of 5×10^5 cells per well and incubated in FBS-free medium for 24 h. On the following day, a scratch was created on the cell monolayer using a 1 mL pipette tip, and the wells were rinsed with PBS to remove suspended cells. A mixed solution of different groups of CM (50%) and endothelial cell medium (50%) was added to the wells. After 12 h of culture, cells were photographed with a light microscope. The scratch width was measured at designated time points and compared to the initial scratch width to calculate the migration rate using the formula:

$$\text{Migration Rate } (\%) = \frac{\text{Scratch Width at timepoint}}{\text{Scratch Width at 0h}} \times 100 \quad (5)$$

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Tube Formation: Chilled Matrigel (Corning, USA) was added to a 48-well plate and kept on ice. To promote polymerization, the plate was incubated at 37 °C for 30 min. HUVECs suspended in a mixture of 50% CM and 50% endothelial cell medium were seeded onto the polymerized Matrigel. After a 2-h incubation, tube formation was imaged, and ImageJ software was used to quantify the total length of the tubes.

ELISA: After culturing RAW264.7 cells in Gel and Gel-Flavonoids for 24 h, the supernatant was collected and diluted with PBS. ELISA kits were employed to detect the concentrations of IL-10 (Beyotime, PI522) and TNF-α (Beyotime, PT512).

Inhibition Test: Sulfo-N-succinimidyl oleate (SSO, 100 μ;M) was utilized as a CD36 inhibitor. Following a day of culture with the CD36 inhibitor, the supernatant of the macrophages was collected for tube formation and ELISA assays.

In Vivo Diabetic Wound Healing Model: The animal study was reviewed and approved by the Peking University Health Centre Institutional Animal Care and Use Committee and Peking University Health Centre Ethics Committee (IACUC number: LA2023018). Healthy 9-week-old SD rats (Charles River) were used to establish the type 1 diabetes model. Rats were injected intraperitoneally with streptozotocin (STZ) solution at a dose of 60 mg k^{-1}g. Rats with blood glucose levels≥ 16.7 mmol L^{-1} after 1 week were considered successful in modeling. An 8-mm full-thickness skin wound was created on the upper back of the rat using a round punch. Wound packs were divided into 8 groups as follows: NC (No hydrogel), 3M, GelMA (Engineering For Life), Gel, Gel-Myr, Gel-2H myr, Gel-Cat, Gel-Que (n = 12). After 3, 7, 14, and 21 days of treatment, the wounds were photographed, and the rats were euthanized by asphyxiation with CO_2. The samples from the wounds were analyzed by histology and immunohistochemistry. Skin specimens were fixed in 4% (w/v) paraformaldehyde for 24 h at room temperature. Then these specimens were embedded in paraffin. Histomorphology staining and analysis were performed on 4μm thickness sections of the central portion of the wound tissue. These sections were prepared for hematoxylin and eosin (H&E) and Masson staining (Solarbio) according to the manufacturer's protocols. For immunohistochemistry detection, skin sections were incubated in primary antibodies overnight at room temperature, including: IL10 and CD31 (1:200, Abcam). The following day sections were incubated with HRP-labeled goat anti-rabbit IgG secondary antibodies (1:200, Abcam). The sections were then stained with 3,3-diaminoamphetamine (DAB) solution. Finally, hematoxylin was stained to reveal the nucleus. Seal the samples with a neutral adhesive. Images were captured using an Olympus microscope.

Statistical Analysis: SPSS 26.0 software was employed for statistical analysis and all charts were generated using GraphPad Prism 9.0. Each test was performed in triplicate and are presented as the mean ± SD. One-way analysis of variance (ANOVA) was used to determine statistical significance. Significance was defined at three levels ($^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

diabetic wound, flavonoids, hydrogel, macrophage, self-healing

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